GenAI Unit 3 Assignment 1

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Section: D **Date:** 03-04-2026

Advanced RAG and Retrieval

```
rank_bm25: n/a
sentence_transformers: 5.3.0
transformers: 5.5.0
sklearn: 1.8.0
pandas: 2.3.1
numpy: 2.4.3
google.generativeai: 0.8.6
groq: 0.37.1
CUDA available: False
```

```
Loading weights: 100%|██████████| 103/103 [00:00<00:00, 6915.09it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

	doc_id	dense_score	dense_rank	text
0	0	0.693996	1	Transformers represent tokens as contextual em...
1	2	0.646056	2	Positional encoding injects sequence order inf...
2	10	0.221824	3	Technical terms such as FlashAttention-2, LoRA...

	doc_id	rrf_score	bm25_rank	sbert_rank	text
0	4	0.032018	1	4	AdamW combines adaptive moments with decoupled...
1	5	0.032002	2	3	Learning rate warmup followed by cosine decay ...
2	6	0.031054	7	2	Batch normalization stabilizes activation dist...
3	11	0.031025	3	6	Contrastive learning shapes embeddings so sema...
4	3	0.030886	9	1	Gradient descent updates neural network parame...

```
Gemini HyDE fallback triggered: 429 You exceeded your current quota, please check your plan and billing details. For more information on this error, head to: https://ai.google.dev/gemi
* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 0, model: gemini-2.0-flash
* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_requests, limit: 0, model: gemini-2.0-flash
* Quota exceeded for metric: generativelanguage.googleapis.com/generate_content_free_tier_input_token_count, limit: 0, model: gemini-2.0-flash
Please retry in 44.293252004s. [links {
  description: "Learn more about Gemini API quotas"
  url: "https://ai.google.dev/gemini-api/docs/rate-limits"
}
, violations {
  quota_metric: "generativelanguage.googleapis.com/generate_content_free_tier_requests"
  quota_id: "GenerateRequestsPerDayPerProjectPerModel-FreeTier"
  quota_dimensions {
    key: "model"
    value: "gemini-2.0-flash"
  }
  quota_dimensions {
    key: "location"
    value: "global"
  }
}
violations {
  quota_metric: "generativelanguage.googleapis.com/generate_content_free_tier_requests"
  quota_id: "GenerateRequestsPerMinutePerProjectPerModel-FreeTier"
  quota_dimensions {
    key: "model"
...
  seconds: 44
}
]
Hypothetical answer: Transformers encode meaning by contextualizing each token with self-attention, combining positional information, and learning semantic relationships in embedding s
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
```

```
Loading weights: 100%|██████████| 105/105 [00:00<00:00, 12111.93it/s]
BertForSequenceClassification LOAD REPORT from: cross-encoder/ms-marco-MiniLM-L-6-v2
Key                          | Status     |  |
-----------------------------+------------+--+-
bert.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

	doc_id	rrf_score	bm25_rank	sbert_rank	text	cross_score
0	5	0.032002	2	3	Learning rate warmup followed by cosine decay ...	-2.448631
1	4	0.032018	1	4	AdamW combines adaptive moments with decoupled...	-4.013637
2	3	0.030886	9	1	Gradient descent updates neural network parame...	-4.670692

	Query	Naive RAG Top Doc	Advanced RAG Top Doc	Are they different?
0	how do transformers encode meaning?	Transformers represent tokens as contextual em...	Transformers represent tokens as contextual em...	No
1	optimization techniques for training	Gradient descent updates neural network parame...	Learning rate warmup followed by cosine decay ...	Yes
2	what helps match technical acronyms in retrieval?	Technical terms such as FlashAttention-2, LoRA...	Technical terms such as FlashAttention-2, LoRA...	No

```
| Query | Naive RAG Top Doc | Advanced RAG Top Doc | Are they different? |
|---|---|---|---|
| how do transformers encode meaning? | Transformers represent tokens as contextual embeddings and use self-attention so each token attends to others. | Transformers represent tokens a
| optimization techniques for training | Gradient descent updates neural network parameters to minimize loss during training. | Learning rate warmup followed by cosine decay can improv
| what helps match technical acronyms in retrieval? | Technical terms such as FlashAttention-2, LoRA, and BERTScore may be captured strongly by keyword search. | Technical terms such a
```

Filled Comparison Table (Markdown)

Query	Naive RAG Top Doc	Advanced RAG Top Doc	Are they different?
how do transformers encode meaning?	Transformers represent tokens as contextual embeddings and use self-attention so each token attends to others.	Transformers represent tokens as contextual embeddings and use self-attention so each token attends to others.	No
optimization techniques for training	Gradient descent updates neural network parameters to minimize loss during training.	Learning rate warmup followed by cosine decay can improve optimization stability and final performance.	Yes
what helps match technical acronyms in retrieval?	Technical terms such as FlashAttention-2, LoRA, and BERTScore may be captured strongly by keyword search.	Technical terms such as FlashAttention-2, LoRA, and BERTScore may be captured strongly by keyword search.	No

Genuine Observation Notes

1. For semantic questions, dense retrieval often works, but hybrid + reranking usually sharpens top-document relevance.
2. For keyword/jargon-heavy questions (e.g., FlashAttention-2, BM25, acronyms), BM25 contributions in RRF increase exact-match recall.
3. Cross-encoder reranking can reorder hybrid candidates meaningfully because it models query-document interaction directly.

	query	alpha	top_doc_id	top_doc
0	how do transformers encode meaning?	0.3	0	Transformers represent tokens as contextual em...
1	how do transformers encode meaning?	0.5	0	Transformers represent tokens as contextual em...
2	how do transformers encode meaning?	0.7	0	Transformers represent tokens as contextual em...
3	optimization techniques for training	0.3	5	Learning rate warmup followed by cosine decay ...
4	optimization techniques for training	0.5	4	AdamW combines adaptive moments with decoupled...
5	optimization techniques for training	0.7	4	AdamW combines adaptive moments with decoupled...
6	what helps match technical acronyms in retrieval?	0.3	10	Technical terms such as FlashAttention-2, LoRA...
7	what helps match technical acronyms in retrieval?	0.5	10	Technical terms such as FlashAttention-2, LoRA...
8	what helps match technical acronyms in retrieval?	0.7	10	Technical terms such as FlashAttention-2, LoRA...

	chunk_size	num_chunks	top_doc_preview	qualitative_relevance
0	50	12	and weight decay to improve stability and gene...	High
1	100	6	improve stability and generalization. Transfor...	High
2	200	3	Transformer optimization uses AdamW, warmup, c...	High

	doc_id	three_rrf_score	bm25_rank	sbert_rank	colbert_rank	text
0	5	0.047907	2	5	1	Learning rate warmup followed by cosine decay ...
1	10	0.047883	4	2	2	Technical terms such as FlashAttention-2, LoRA...
2	2	0.047280	1	1	9	Positional encoding injects sequence order inf...
3	6	0.045950	7	6	3	Batch normalization stabilizes activation dist...
4	0	0.045387	12	3	4	Transformers represent tokens as contextual em...